Jason A. Bott

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Curriculum Vitae

EDUCATION

2022 - 2026 ***B.S.*** **Geophysics** ***Honors***
The University of Texas at Austin Jackson School of Geosciences
Thesis: *"Evidence for Subglacial Volcanism Beneath Thwaites Glacier from Ice Penetrating Radar"*

2024 - 2026 **Certificate in Computational Science and Engineering**
UT Austin Oden Institute for Computational Science and Engineering
Indep. Research Project: *"Examining relationships between the geomorphological evolution of the Amazon River Basin and spatial variance in the birdsong of the family Thamnophilidae*"

RESEARCH & PROFESSIONAL EXPERIENCE

Undergraduate Researcher — *University of Texas College of Natural Sciences & Oden Institute for Computational Engineering*
Aug 2026 – Present
Advisor: Dr. Timothy Keitt

- Leading an independent research project using an open-source machine learning tool (BirdNET) to examine birdsong variance in the antbird family (*Thamnophilidae*) within geographic space for population assessment and to examine their relationship with the geomorphological evolution of local rivers.

Undergraduate Research Associate — *University of Texas Institute for Geophysics (UTIG)*
Nov 2022 – Present
Advisors: Drs. Don Blankenship, Duncan Young, and Shuai Yan

- Used ice-penetrating-radar (IPR) coupled with numerical ice models to conduct research on a multitude of glaciological settings such as the Devon Ice Cap, Canada and Thwaites Glacier, Antarctica.
- Led research on the potential effects of subglacial volcanism on shear margin mechanics in the Thwaites Glacier catchment, combining IPR data with thermally-coupled ice-flow models (shallow ice approximation and 3D finite-difference Stokes-flow).
- Led research comparing geospatial interpolation techniques (ordinary kriging, stochastic simulation of subglacial bed roughness, mass-balance-model-informed topography, etc.) for subglacial digital elevation model (DEM) generation, and quantified their downstream effects on shallow ice approximation model outputs.
- Presented research at international conferences and invited institute talks.

Field Science/Engineering Assistant — *University of Texas Institute for Geophysics (UTIG) & Korean Polar Research Institute*
Dec 2025 – Feb 2026

- Served as a field technician on UTIG's 4-person Airborne Ice-Penetrating-Radar (IPR) team during an international Antarctic expedition to the Amundsen Sea Embayment aboard the RV Araon icebreaker.
- Planned helicopter radar surveys, assisted Senior Radar Engineer in operation and troubleshooting of IPR instrumentation in the field, and processed/QC'd radar data in real time using Python and Linux.
- Contributed to record-breaking inland radar coverage, novel measurements of ice crystal-fabric-orientation, and the first-ever measurements of previously unmapped subglacial terrain.

Technical Project Lead — *DOE Geothermal Collegiate Competition*
Sep 2024 – Dec 2024
Advisor: Dr. Malcolm Ross

- Led a 6-person Applied Science & Engineering team in the U.S. Department of Energy Geothermal Collegiate

Competition.
- Coordinated interdisciplinary technical work across the team to propose an economically viable dual-use carbon capturing greenhouse complex in Presidio County, Texas, powered by direct-use geothermal energy.

Founder, Host & Producer — *On the Subject of Life* Podcast*
Oct 2023–Jan 2025

- Produced and hosted deeply researched, long-form (>2 hour) video interviews with scientists and domain experts.
- Managed end-to-end production, research, and outreach.
- 19+ free-to-watch episodes with guests such as Dr. Dan Stanzione (UT Austin's Associate Vice President for Research) and Dr. Michael Pyrcz (Geostatistics Expert and Tenured Professor at UT Austin).

GRANTS & FELLOWSHIPS

- University of Texas Undergraduate Research Fellowship — **$600** (Mar 2025)
- University of Texas Undergraduate Research Travel Fund — **$300** (Mar 2025)

HONORS & AWARDS

- Louis and Elizabeth Scherck Geology Scholarship Recipient (2022-2025)
- UT Undergraduate Research Fellowship Recipient (2025)
- Jackson School of Geosciences Undergraduate Honors Research Program (2024–Present)
- Dell Scholar, UT Austin (2022–Present)
- Jackson Scholar, UT Austin (2022–Present)
- National Hispanic Scholar (2021)
- Texas 6A District & Area Medalist, Track & Field (2021)

CONFERENCE ABSTRACTS

Bott, J., Blankenship, D., Young, D., Yan, S. (2025). *Why assumptions and uncertainties in bed DEMs matter: A geothermal example from Haynes Glacier, West Antarctica.* EGU General Assembly, Vienna, Austria. https://doi.org/10.5194/egusphere-egu25-14386

Bott, J., Blankenship, D. D., Yan, S., Beem, L., Young, D. A. (2024). *The potential role of anomalous geothermal flux for enhanced basal melting and suppressed ice velocity at Haynes Glacier, West Antarctica.* EGU.

Bott, J., Blankenship, D. D., Yan, S., Young, D. A. (2023). *Potential sources of a velocity anomaly at the grounding line of Haynes Glacier.* AGU Fall Meeting.

Yan, S., Pierce, C., Dapeng, Z., Czwakiel, N., Jackson, H., **Bott, J.**, Blankenship, D. (2026). *On the possibility of a substantial but ephemeral salty lake beneath the Laurentide Ice Sheet.* AGU General Assembly.

D. Li, D.D. Blankenship, D. Young, S. Yan, D. Buhl, C. Pierce, L. Beem, J. Houghton, **J. Bott**, C.K. Lee, K. Mankoff, E. Quartini, S. Kempf, G. Ng, W.S. Lee (2026). *Radar-Derived Ice Fabric Anisotropy and Anisotropic Scattering in Thwaites Glacier, West Antarctica.* IGS Radioglaciology.

Gerekos, C., Rutishauser, A., Scanlan, K., Wolfenbarger, N., Beem, L., **Bott, J.**, Blankenship, D. (2023). *Airborne radar radiometry and coastline mapping of specular subglacial terrain on Devon Island.* EGU. https://doi.org/10.5194/egusphere-egu23-9149

TALKS & LECTURES

UTIG Discussion Hour — *Why uncertainties and assumptions in bed DEMs matter: A geothermal example from the Haynes and Thwaites Glaciers of West Antarctica* (Apr 8, 2025)

UTIG Discussion Hour — *Ice and fire at the world's edge: Anomalous geothermal flux beneath the West Antarctic Ice Sheet and implications for ice-sheet stability* (Oct 23, 2024)

MEDIA COVERAGE

PBS NewsHour, & O'brien, M. (2026, February 12). Despite setback, researchers uncover new findings at Antarctica's Thwaites Glacier. YouTube. https://www.youtube.com/watch?v=T5KnkhG_obc [Coverage of UTIG's Thwaites Glacier radar survey field operations]

Zhong, R. & The New York Times. (2026, January 27). Scanning Ice for Monster Cracks, on This World and Another. The New York Times. https://www.nytimes.com/2026/01/27/climate/antarctica-thwaites-radar.html [Featured/Quoted]

Thakur, M. (2024, June 28). *UT student hosts video podcast featuring conversations with expert guests*. The Daily Texan. [Featured/Quoted]
https://thedailytexan.com/2024/06/28/ut-student-hosts-video-podcast-featuring-conversations-with-expert-guests/

RELEVANT COURSEWORK

Geosciences

- GEO365P Potential Fields in Geophysics ..... A-
- GEO325K Computational Methods in Geophysics ..... A-
- GEO325M Continuum Mechanics ..... A
- GEO325C Numerical Modeling in Geoscience ..... A
- GEO428 Structural Geology ..... A-
- GEO348K Marine Geology & Geophysics ..... A

Applied Mathematics

- COE379L Introduction to Machine Learning ..... A
- COE391 Inverse Theory & Variational Calculus ..... A
- PGE338 Geostatistics & Data Analysis ..... A
- PGE311 Numerical Methods & Programming ..... A

Mathematics (M)

- M427J Differential Equations & Linear Algebra ..... A-
- M427L Advanced Calculus for Applications ..... A

Foreign Language

- JPN601D Japanese I ..... A-

FIELDWORK

- KOPRI Thwaites Glacier Cruise, Amundsen Sea Embayment, West Antarctica (Dec 2025–Feb 2026)
- Structural Geology Mapping Field Trip — Big Bend National Park, TX (Nov 2025)
- Marine Geology & Geophysics Field Course — Port Aransas, TX (May 2024)
- Marine Sediment Mapping & Sea-Level Reconstruction — San Salvador Island, the Bahamas (Feb–Mar 2024)

- Introductory Geoscience Field Course — Austin, TX (Aug-Dec 2022)

PROFESSIONAL MEMBERSHIPS

- European Geosciences Union (EGU)
- American Geophysical Union (AGU)

CERTIFICATIONS & SAFETY TRAINING

- STCW Basic Firefighting (MARTSA-53)
- STCW Adult First Aid/CPR/AED
- STCW Personal Survival Techniques
- STCW Personal Safety & Social Responsibilities (MARTSA-359)

LANGUAGES

- English — Native
- Spanish, Español — Intermediate
- Japanese, 日本語 — N4
- Indonesian, Bahasa Indonesia — Novice

LINKS

- Personal Website: https://portfolio-jasonbott.squarespace.com/
- Research Profile: https://ig.utexas.edu/students/jason-bott/
- Google Scholar: https://scholar.google.com/citations?user=31Ae8UkAAAAJ
- Podcast: www.heightsofhumanitypodcast.com

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